

01_Fundamentals — Folder Index

PostgreSQL Complete Guide

Module 1: Database Fundamentals

Prerequisites: None | Difficulty: Beginner → Intermediate

Overview

This module builds the conceptual foundation required before writing a single line of SQL. Understanding *why* databases exist, *what* makes PostgreSQL different, and *how* fundamental principles like ACID and CAP apply to real systems will make you a far better engineer than memorizing syntax alone.

Chapters in This Module

#	File	Topic	Difficulty	Time
01	01_what_is_a_database.md	What is a Database?	Beginner	20 min
02	02_types_of_databases.md	Types of Databases	Beginner	25 min
03	03_relational_databases.md	Relational Databases	Beginner-Intermediate	30 min
04	04_oltp_vs_olap.md	OLTP vs. OLAP	Intermediate	25 min
05	05_cap_theorem.md	CAP Theorem	Intermediate	25 min
06	06_acid_properties.md	ACID Properties	Intermediate	30 min
07	07_base_properties.md	BASE Properties	Intermediate	20 min
08	08_postgresql_introduction.md	PostgreSQL Introduction	Beginner	25 min
09	09_installation_guide.md	Installation Guide	Beginner	45 min

Total Estimated Time: ~3.5 hours

Learning Path

Start Here

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v

01_what_is_a_database --> Understand what a database IS

|

```
    v
02_types_of_databases --> Know the landscape (SQL, NoSQL, NewSQL)
    |
    v
03_relational_databases --> Master the relational model (keys, joins, normalization)
    |
    v
04_oltp_vs_olap         --> Understand workload types and schema design implications
    |
    v
05_cap_theorem          --> Grasp distributed systems trade-offs
    |
    v
06_acid_properties      --> Understand transaction guarantees deeply (critical!)
    |
    v
07_base_properties      --> See the other side: eventual consistency
    |
    v
08_postgresql_introduction --> Meet your database engine
    |
    v
09_installation_guide   --> Get PostgreSQL running on your machine
    |
    v
02_SQL_Basics/ --> Next module: Start writing SQL!
```

Key Concepts Covered

Database Fundamentals

- What is a database vs. a DBMS vs. a database system
- Problems databases solve over flat-file storage
- The three-schema architecture (external/conceptual/internal)

Data Models

- Relational, document, key-value, wide-column, graph, time-series
- When to use each type
- Polyglot persistence

The Relational Model

- Tables, rows, columns, domains
- Primary keys, foreign keys, candidate keys, composite keys
- One-to-one, one-to-many, many-to-many relationships
- First, Second, Third Normal Form
- Referential integrity and constraints

Workload Types

- OLTP: transactional, normalized, row-oriented
- OLAP: analytical, denormalized, star schema
- ETL pipelines and data warehousing

Distributed Systems Theory

- CAP Theorem: Consistency, Availability, Partition Tolerance
- PACELC model
- Consistency models: linearizable → eventual
- ACID: Atomicity, Consistency, Isolation, Durability
- BASE: Basically Available, Soft State, Eventually Consistent
- MVCC and transaction isolation levels

PostgreSQL Specifics

- History (INGRES → POSTGRES → PostgreSQL)
 - Process architecture (multi-process, shared memory)
 - Key features: JSONB, arrays, extensions, custom types
 - Installation on all major platforms
 - Post-installation security and configuration
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Quick Reference: Key SQL from This Module

```
-- Connect to PostgreSQL
psql -U postgres -d mydb

-- Create your first database
CREATE DATABASE learning_db;

-- Create a normalized table
CREATE TABLE customers (
    customer_id SERIAL PRIMARY KEY,
    email        VARCHAR(100) UNIQUE NOT NULL,
    created_at   TIMESTAMPTZ DEFAULT NOW()
);

-- Check ACID in action
BEGIN;
    UPDATE accounts SET balance = balance - 100 WHERE id = 1;
    UPDATE accounts SET balance = balance + 100 WHERE id = 2;
COMMIT;

-- MVCC: Multiple isolation levels
BEGIN TRANSACTION ISOLATION LEVEL REPEATABLE READ;
    SELECT * FROM products WHERE id = 1;
COMMIT;

-- Check replication lag (distributed systems)
SELECT now() - pg_last_xact_replay_timestamp() AS lag;
```

Interview Topics From This Module

These are commonly asked in database/backend engineering interviews:

1. What problems do databases solve that flat files cannot?
 2. What is the relational model? Who invented it?
 3. What are the ACID properties?
 4. What is the difference between ACID-Consistency and CAP-Consistency?
 5. What are the four transaction isolation levels?
 6. What read anomalies does each isolation level prevent?
 7. What is MVCC and how does it enable high concurrency?
 8. What is the CAP theorem?
 9. Why can't a distributed system guarantee all three CAP properties?
 10. What is the difference between OLTP and OLAP workloads?
 11. What is a star schema?
 12. What is eventual consistency?
 13. What is the difference between BASE and ACID?
 14. What makes PostgreSQL different from MySQL?
 15. How does PostgreSQL implement durability?
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Prerequisites for Next Module

After completing this module, you should be able to:

- Connect to a PostgreSQL instance using `psql`
- Create databases and basic tables
- Understand why transactions matter
- Explain ACID properties in an interview
- Know when to choose PostgreSQL vs. other database types

Next: [02 SQL Basics/README.md](#) — DDL, DML, SELECT, filtering, functions

Resources

Official Documentation

- [PostgreSQL Documentation](#)
- [PostgreSQL Wiki](#)

Books Referenced

- *An Introduction to Database Systems* — C.J. Date
- *PostgreSQL: Up and Running* — Regina Obe & Leo Hsu
- *The Art of PostgreSQL* — Dimitri Fontaine
- *Designing Data-Intensive Applications* — Martin Kleppmann (for CAP/distributed systems)

Online Tools

- [DB-Engines Ranking](#)
 - [use-the-index-luke.com](#) — SQL indexing explained
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Module 1 of the PostgreSQL Complete Guide | Author: Sudhir Jangra